Splitting Agree to [φ] and [INDEX]: Evidence from Chinese Reflexives

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Overview

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Reflexive binding is commonly derived by valuing an anaphor's deficient [φ]-features through Agree with a higher antecedent (Reuland 2006; Heinat 2006; Kratzer 2009). Yet [φ]-feature valuation alone does not explain semantic co-indexation; I argue that [φ]-features and the Binding Index [BI] must be distinguished in the grammar:

- ☞ Empirically, this study provides a unified account of Long-distance Binding and the person-sensitive Blocking Effect in Mandarin Chinese within a Cyclic Spell-out framework, providing an important typological insight into how *SE*-type anaphor and *SELF*-type anaphor may coexist and interact within a single grammar.
- ☞ Theoretically, this study revitalizes the syntactic approach to reflexive binding by locating the source of semantic co-indexation in syntactic computation. The two split Agree operations, $\text{Agree}_{[*\varphi:_*]}$ and $\text{Agree}_{[*BI:_*]}$, differ in their properties, timing, and interactions with Internal Merge and Cyclic Spell-out, which derives the puzzled empirical patterns and their opacity issues of Chinese reflexives.

Empirical Data

Two Chinese Reflexives

Mandarin Chinese features two distinct types of reflexive constructions:

☞ Bare Reflexive: *SE*-anaphor

ziji

self

☞ Compound Reflexive: *SELF*-anaphor

wo/ni/ta-ziji

1/2/3SG-self

(0a) Zhangsan$_i$ xihuan ziji$_i$.
John like self
'John$_i$ likes himself$_i$.'

(0b) Zhangsan$_i$ xihuan ta-ziji$_i$.
John like 3SG-self
'John$_i$ likes himself$_i$.'

Empirical Puzzles

Only bare *ziji* permits **long-distance binding**, whereas the complex form is strictly local (Battistella & Xu 1990; Cole et al. 1990; Huang & Tang 1991; Xu 1993; Pan 2001; Huang &Liu 2001; among others).[1]

(1) a. Zhangsan$_i$ renwei [Lisi$_j$ zhidao [Wangwu$_k$ xihuan ziji$_{i/j/k}$]].
John think Lee know Wayne like self
'John$_i$ thinks that Lee$_j$ knows that Wayne$_k$ likes himself$_{i/j/k}$.'

b. Zhangsan$_i$ renwei [Lisi$_j$ zhidao [Wangwu$_k$ xihuan ta-ziji$_{*i/*j/k}$]].
John think Lee know Wayne like 3SG-self
'John$_i$ thinks that Lee$_j$ knows that Wayne$_k$ likes himself$_{*i/*j/k}$.'

(Cole & Sung, 1994, p. 357)

[1] A substantial body of literature has shown that the verb *renwei* ('think') selects a genuine CP, using diagnostics such as embedded topicalization and compatibility with modal verbs.

Empirical Puzzles

Additionally, *ziji* exhibits a person-sensitive **Blocking Effect**: its long-distance binding requires person agreement among the relevant subjects, with any person mismatch between the intermediate DP (embedded Subj.) and the long-distance DP (matrix Subj.) blocking the dependency (Battistella & Xu 1990; Cole et al. 1990; Huang & Tang 1991; Xu 1993; Pan 2001; Huang &Liu 2001; among others)[2]:

(2) a. Zhangsan$_i$ renwei [wo$_j$/ni$_k$ hui zebei ziji$_{*i/j/k}$].
John think 1SG/2SG will blame self
'John$_i$ thinks that I$_j$/you$_k$ will blame *himself$_i$/ ✓myself$_j$/ ✓yourself$_k$.'

b. Wo$_j$/Ni$_k$ renwei [Zhangsan$_i$ hui zebei ziji$_{i/*j/*k}$].
1SG/2SG think John will blame self
'I$_j$/you$_k$ think that John$_i$ will blame ✓himself$_i$/ *myself$_j$/ *yourself$_k$.'

[2] Among all the literature, some, like Pan 1997, 2001, have different judgment of (2b), arguing that the 3SG is not a necessary blocker like 1SG and 2SG.

Previous Analysis

Semantic Approach

The functional approach treats *ziji* as a Logophoric Reflexive anchored to a perspective center (Pan 2000, 2013; Huang & Liu 2001; Oshima 2006, 2007; Wang & Pan 2015; Charnavel et al. 2017; a.o.). Long-distance binding reflects alignment with the matrix attitude holder, while blocking arises when first- or second-person interveners introduce a competing discourse center ("anchor" in Huang 1984).

(3) a. Zhangsan$_i$ renwei [Lisi$_j$ chang piping ziji$_{i/j}$].
John think Lee often criticize self
'John$_i$ thinks that Lee$_j$ often criticizes himself$_{i/j}$.'

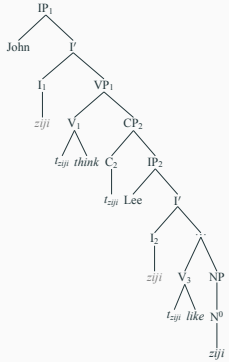
b. Zhangsan$_i$ renwei: "Lisi$_j$ chang piping wo$_{i/*j}$".
John think Lee often criticize 1SG
'John$_i$ thinks: "Lee$_j$ often criticizes me$_{i/*j}$".'

(Huang & Liu, 2001, p. 156)

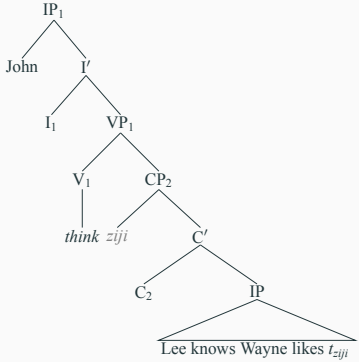
☞ Problem: How should we explain the Monomorphemic Effect???

Syntactic Approach

The formal tradition treats long-distance binding as a phenomenon derivable from local binding through successive movement at Logical Form (LF) (Tang 1989; Battistella 1989; Cole et al. 1990; Huang & Tang 1991 Cole & Sung 1994; Cole & Wang 1996; Huang & Liu 2001 a.o.).



Head-movement approach



CP-adjunction approach

Syntactic Approach

The Blocking Effect follows naturally under these two approaches: having acquired φ-features from the local antecedent before LF movement, the bare *ziji* can only take a long-distance antecedent with the same φ-features as the local one. However, what about the Monomorphemic Effect?

☞ **Head-movement Approach:**

> *"The requirement of monomorphemicity follows from the natural assumption that only monomorphemic ziji is an X^0 category which can undergo head movement, but not polymorphemic, phrasal ta-ziji, etc."*

(Huang & Liu 2001 on Cole, Hermon, Sung 1990; Cole & Sung 1994; Cole & Wang 1996; a.o.)

☞ **CP/IP-adjunction Approach:**

Reflexives have [φ-Index] and [*R-Index*]:

- ☞ Bare *ziji* → [*R-Index*] licensed at LF
- ☞ Compound *wo/ni/ta-ziji* → [*R-Index*] licensed at Surface Structure (SS)

(Huang & Tang 1991)

Licensing Reflexives

Two Types of Agree Operations

If we follow Reuland (2001, 2006, 2022) that establishing an anaphoric relation between a reflexive pronoun and its antecedent requires the reflexive to enter into an Agree relation, the question to be answered is: **what should be agreed with?**

☞ [φ-features] (Heinat 2006, 2009; Reuland 2006)

☞ [Binding Index] (Adger & Ramchand 2005; Hicks 2009)

☞ [φ-features] AND [Binding Index] (Rezac 2006; Murphy & Meyase 2022)

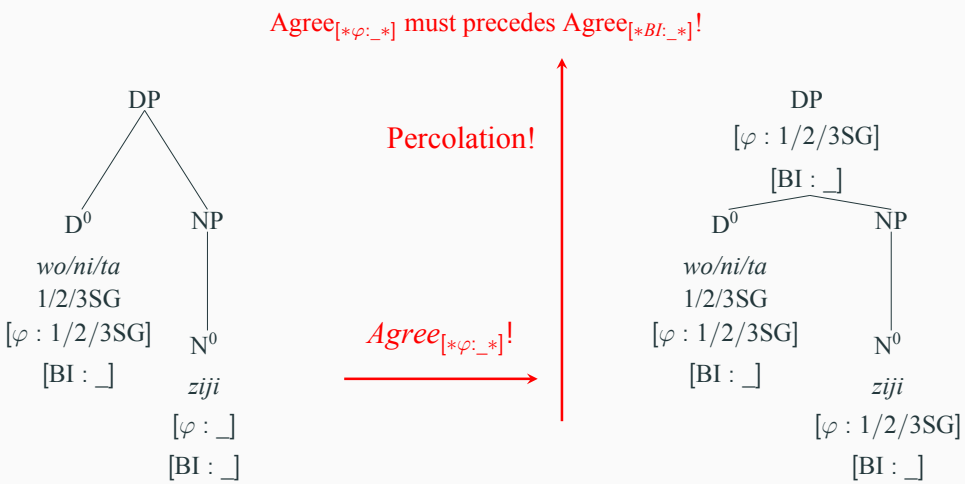
Index valuation proceeds in parallel with φ-feature valuation, essentially as a by-product governed by Chomsky's (1995) Free Rider Principle:

$$[_{HP}\ DP_{[\varphi:n][ix=m]}\ \ldots\ Ref_{[\varphi:_][ix=_]}\ \ldots] \xrightarrow{\text{Agree!}} [_{HP}\ DP_{[\varphi:n][ix=m]}\ \ldots\ Ref_{[\varphi:n][ix=m]}\ \ldots]$$

(Rezac, 2006)

However, this does not apply for Chinese data!

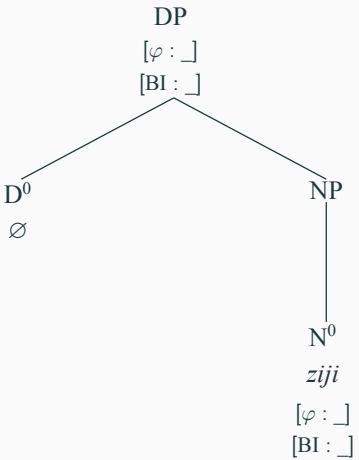
Internal Syntax of Compound Reflexives



(Structure followed Reinhart & Reuland (1993) and Reuland (2001))

Internal Syntax of Bare Reflexives

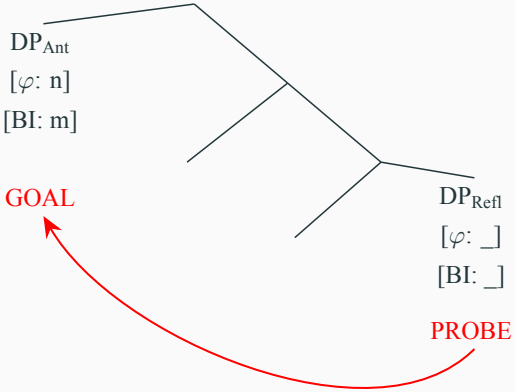
Bare *ziji* also projects a DP, however, with an empty D head:



Upward Agree

In which fashion should agree apply?

Following Müller (2025), the reflexive itself acts as the Probe and searches upward for a c-commanding potential antecedent as its Goal.

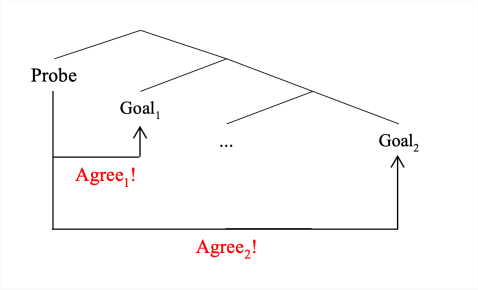


Multiple Agree and Cyclic Agree

Multiple Agree

Hiraiwa (2015)

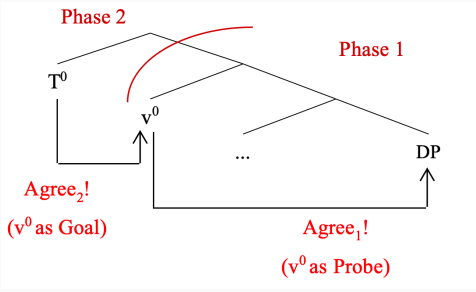
"AGREE applies to all the matched goals at the same derivational point derivationally simultaneously."



Cyclic Agree

Legate (2005)

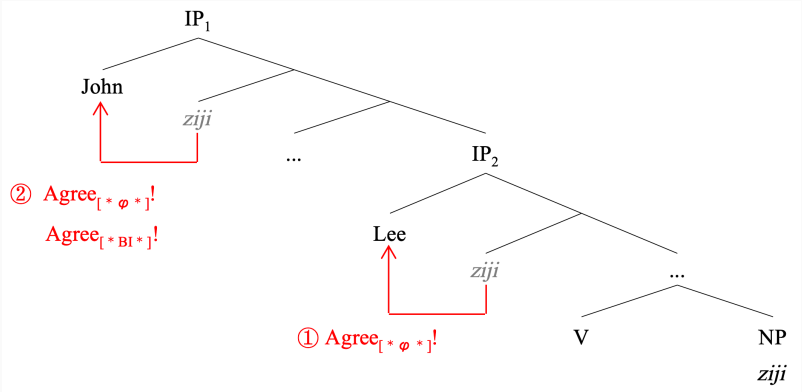
Agree proceeds phase by phase cyclically, with each intervening phase head first probing for the relevant feature and then serving as the goal of the next higher probe.



Multiple Cyclic Agree

☞ Long-distance binding of bare *ziji* requires $\text{Agree}_{[*BI:_*]}$ to be iterable and optional, motivating Multiple Agree! **But not simultaneously!**

☞ Blocking effect of bare *ziji* requires φ-feature matching among all Goals in the $\text{Agree}_{[*\varphi:_*]}$ chain, motivating Cyclic Agree! **But invariable Probe!**



Proposal:
A Last-resort Movement Approach

Key Assumptions

☞ Every step of Agree must be local via formal feature-checking operations! (Cole et al. 1990; Huang & Liu 2001)

☞ Phase Impenetrability Condition (PIC): vP, CP and DP is phases! (Chomsky 2001, 2008; Bošković 2012, 2013, 2014)

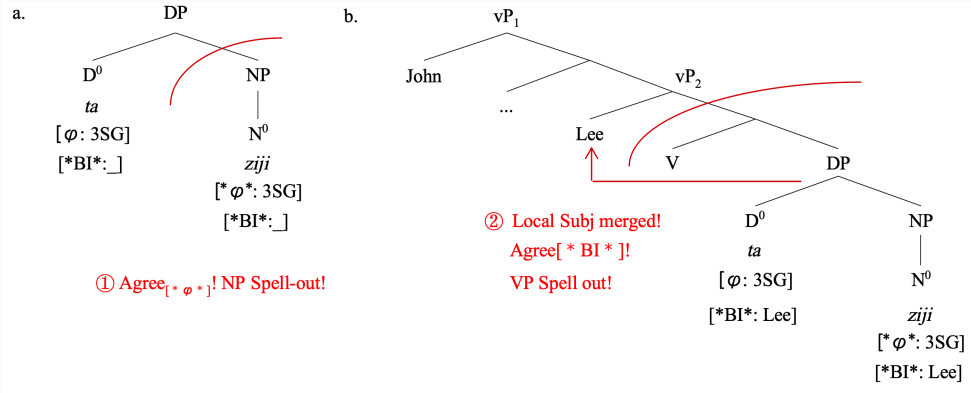
☞ Movement of *ziji* must be successive-cyclic at narrow syntax, instead of LF!

☞ *Ziji* is obligatorily realized as the lower copy (technical stipulation)!

☞ $\text{Agree}_{[*\varphi:_*]}$, bearing a **strong satisfaction condition**, must be valued immediately and can apply only once; once valued, it triggers Spell-Out of the DP phase; whereas $\text{Agree}_{[*BI:_*]}$ can apply at a later point in the derivation and is free to apply at different stages, even after the relevant DP has undergone Spell-Out!

Deriving Compound Reflexive ta-ziji (only local binding)

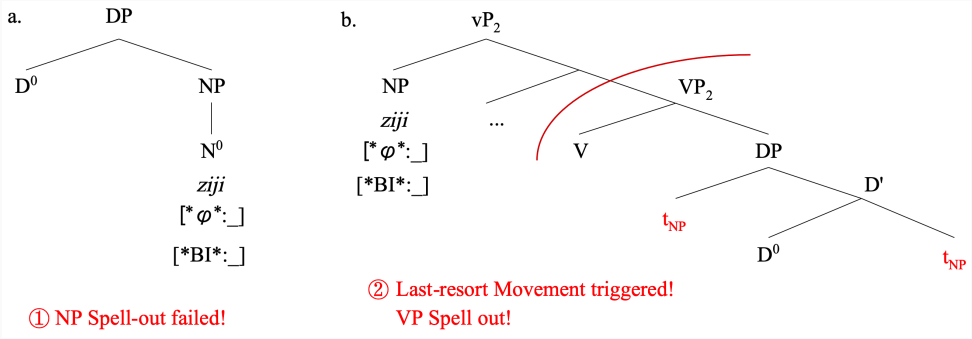
Both DP reflexive *ta-ziji* is frozen in situ and can no longer undergo movement to establish a referential relation via Agree with a higher antecedent, thus deriving Monomorphemic Effect:[3]



[3] $\text{Agree}_{[*BI:_*]}$ with local subject is also local because the local subject agrees with the whole DP and then/simultaneously percolate BI-feature into N head *ziji*.

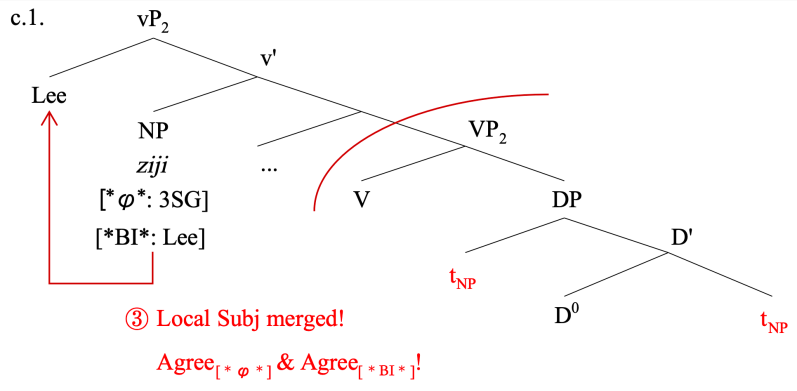
Deriving bare Reflexive ziji

Agree$_{[*\varphi:_*]}$ cannot be satisfied immediately upon Merge, leading to a failure of NP Spell-out. Crucially, this derivational failure forces *ziji* to undergo successive-cyclic movement as a Last Resort strategy to the edge of each phase:



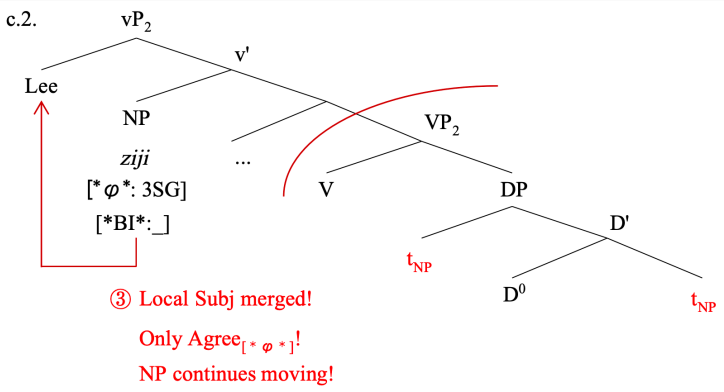
Deriving bare Reflexive ziji (local binding)

Agree$_{[*BI:_*]}$ may either apply immediately, yielding local binding:



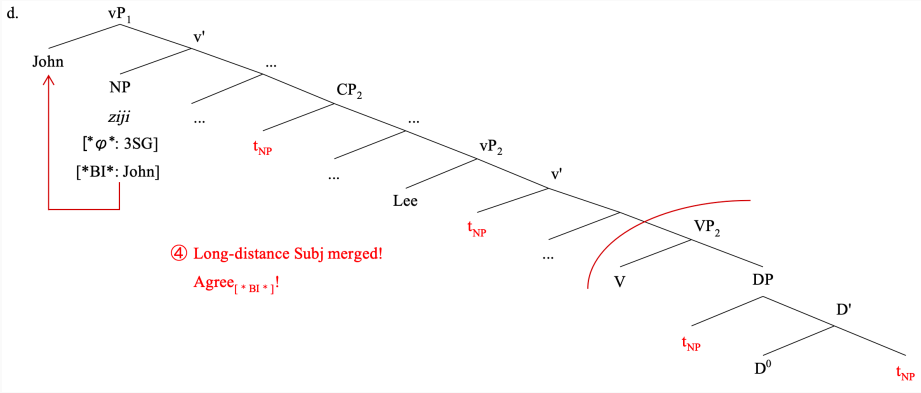
Deriving bare Reflexive ziji (long-distance binding)

Agree$_{[*BI:_*]}$ may also be delayed, but at this intermediate landing site, as soon as the local subject is merged, Agree$_{[*\varphi:_*]}$ must apply and remains unchangeable (cf. Pesetsky (1989)'s Earliness Principle):



Deriving bare Reflexive ziji (long-distance binding)

If *ziji* moves successive-cyclically from the phase edge to the matrix clause, and φ-feature matches, Agree$_{[*BI:_*]}$ takes place with a higher antecedent, thus deriving long-distance binding:



Blocking Effect

The present analysis, like all previous formal analysis, attributes the Blocking Effect to φ-feature mismatch.

However, in sharp contrast to Huang & Tang (1991), the current analysis assumes that NP-movement and Agree operate independently, namely, the failure of Agree does not lead to the impossibility of subsequent head movement, correctly predicting the following data:

(4) Zhangsan$_i$ yiwei [wo$_j$ zhidao [Lisi$_k$ bu-xihuan ziji$_{i/*j/k}$]].
John think 1SG know Lee Neg-like self

'John$_i$ thinks that I$_j$'ve known that Lee$_k$ doesn't like ✓himself$_i$/ *myself$_j$/ ✓himself$_k$.'

Conclusion

Conclusion

By splitting the Agree operation into Agree$_{[*\varphi:_*]}$ and Agree$_{[*BI:_*]}$ under a Cyclic Spell-out approach, this study offers a novel explanation for the different syntactic behaviors of the two reflexive types:

☞ **Monomorphemic Effect**: the absence of the overt [φ]-bearing D head in the bare *ziji* leads to the failure of DP Spell-out, giving rise to the last-resort movement and thus counter-bleeding the long-distance reading, which is bled by the too-early Spelled-out compound DP;

☞ **Blocking Effect**: Person-matching is always enforced in the semantics, as a result of Referential Identity (Sundaresan 2020).

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